

A PRIME-REPRESENTING FUNCTION

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A function $f(x)$ is said to be a prime-representing function if $f(x)$ is a prime number for all positive integral values of x . It will be shown that there exists a real number A such that $[A^{3^x}]$ is a prime-representing function, where $[R]$ denotes the greatest integer less than or equal to R .

Let p_n denote the n th prime number. A. E. Ingham¹ has shown that

$$(1) \quad p_{n+1} - p_n < K p_n^{5/8}$$

where K is a fixed positive integer.

Lemma. *If N is an integer greater than K^3 there exists a prime p such that $N^3 < p < (N+1)^3 - 1$.*

Proof. Let p_n be the greatest prime less than N^3 . Then

$$(2) \quad \begin{aligned} N^3 < p_{n+1} < p_n + K p_n^{5/8} < N^3 + K N^{15/8} < N^3 + N^2 \\ < (N+1)^3 - 1. \end{aligned}$$

Let P_0 be a prime greater than K^3 . Then by the lemma we can construct an infinite sequence of primes, $P_0, P_1, P_2, \dots$, such that $P_n^3 < P_{n+1} < (P_n + 1)^3 - 1$. Let

$$(3) \quad u_n = P_n^{3-n}, \quad v_n = (P_n + 1)^{3-n}.$$

Then

$$(4) \quad v_n > u_n, \quad u_{n+1} = P_{n+1}^{3-n-1} > P_n^{3-n} = u_n,$$

$$(5) \quad v_{n+1} = (P_{n+1} + 1)^{3-n-1} < (P_n + 1)^{3-n} = v_n.$$

It follows at once that the u_n form a bounded monotone increasing sequence. Let $A = \lim_{n \rightarrow \infty} u_n$.

Theorem. $[A^{3^x}]$ is a prime-representing function.

Proof. From (4) and (5) it follows that $u_n < A < v_n$, or $P_n < A^{3^n} < P_n + 1$. Therefore $[A^{3^n}] = P_n$ and $[A^{3^x}]$ is a prime-representing function.

¹ A. E. Ingham, *On the difference between consecutive primes*, Quart. J. Math. Oxford Ser. vol. 8 (1937) pp. 255-266.